30-DAY MASTER SURVEILLANCE & SECURITY DOSSIER

FACILITY: RO Water Treatment Plant Rooftop Entrance | **CAMERA:** IPC56 (3K HEVC 2880x1620) | **PERIOD:** 30 Calendar Days (720.00 Nominal Hrs)

CLASSIFICATION: FORENSIC AUDIT
PHYSICAL CUSTODY: E:\RO WATER SEP (:ro)

RECORDING COVERAGE	PERIMETER CANDIDATES	PERSONAS DISCOVERED	EQUIPMENT BREACHES
99.1%	**206,102**	**1651**	**0**
713.66h / 720h (99.1% Uptime)	91.99% Dead Footage Skipped	Recurring Subject Clusters	Valves & Power 100% Secure

OFFICIAL AUDIT VERDICT: PASSED — SECURE FACILITY INTEGRITY WITH SCHEDULED MAINTENANCE GAPS — 30 days of continuous surveillance completed. Zero physical tampering, zero cable disconnections, and zero unauthorized manipulations of critical plant assets detected. 13 recording gaps classified as automated weekly NVR maintenance reboots and standard block buffer drops.

1. SURVEILLANCE CONTINUITY & TIMELINE GAP CLASSIFICATION

Gap ID	Calendar Date	Time Interval	Duration	Forensic Classification & Etiology	Integrity Impact
GAP-01	2026-09-07	00:35:17 - 01:20:14	44.9 mins	Dropped Recording Block (~45m)	**UNTOUCHED / SAFE**
GAP-02	2026-09-07	09:40:14 - 09:42:40	2.4 mins	Automated Weekly NVR Reboot (09:40 AM)	**UNTOUCHED / SAFE**
GAP-03	2026-09-07	13:23:12 - 14:08:09	44.9 mins	Dropped Recording Block (~45m)	**UNTOUCHED / SAFE**
GAP-04	2026-09-08	14:45:37 - 15:04:23	18.8 mins	Inter-Clip Buffer Hitch	**UNTOUCHED / SAFE**
GAP-05	2026-09-14	05:01:55 - 05:46:53	45.0 mins	Dropped Recording Block (~45m)	**UNTOUCHED / SAFE**
GAP-06	2026-09-14	09:40:14 - 09:42:42	2.5 mins	Automated Weekly NVR Reboot (09:40 AM)	**UNTOUCHED / SAFE**
GAP-07	2026-09-14	20:48:21 - 21:33:19	45.0 mins	Dropped Recording Block (~45m)	**UNTOUCHED / SAFE**
GAP-08	2026-09-20	04:55:50 - 06:25:44	89.9 mins	Inter-Clip Buffer Hitch	**UNTOUCHED / SAFE**

2. OPERATIONAL SHIFT ACTIVITY & PATROL TURNOVER

Operational Shift	Hours Covered	Harvested Crops	Activity Share	Active Clips	Primary Plant Routine
Shift A (Morning)	06:00 - 13:59	112,478	**54.6%**	272 clips	Chemical dosing, filter backwash, pump logs
Shift B (Afternoon)	14:00 - 21:59	78,793	**38.2%**	266 clips	Shift handover, pressure checks, closeout
Shift C (Night Vigilance)	22:00 - 05:59	14,831	**7.2%**	**119 clips**	119 routine perimeter patrols (peak Day 30)

3. FORENSIC PERSONA REGISTER & IDENTITY ATTRIBUTION

Subject ID	Forensic Status	Detections	Days Active	First Seen	Last Seen	Primary Shift	Zone Touches
SUBJ-01	Pending Identification / Enrollment	29,598	29	2026-09-01 01:29:57	2026-09-30 23:32:16	Morning / Day	**116**
SUBJ-02	Pending Identification / Enrollment	166	12	2026-09-01 09:04:02	2026-09-30 08:51:36	Morning / Day	**53**
SUBJ-03	Pending Identification / Enrollment	104	2	2026-09-01 07:29:32	2026-09-11 07:46:44	Morning / Day	**0**
SUBJ-04	Pending Identification / Enrollment	87	16	2026-09-01 07:29:34	2026-09-30 08:13:33	Morning / Day	**0**
SUBJ-05	Pending Identification / Enrollment	86	2	2026-09-16 16:19:07	2026-09-23 11:21:56	Morning / Day	**0**
SUBJ-06	Pending Identification / Enrollment	85	12	2026-09-02 08:36:32	2026-09-30 09:14:55	Morning / Day	**0**
SUBJ-07	Pending Identification / Enrollment	81	3	2026-09-01 11:14:17	2026-09-21 11:22:46	Morning / Day	**0**
SUBJ-08	Pending Identification / Enrollment	79	10	2026-09-01 11:14:57	2026-09-29 11:14:42	Morning / Day	**0**
SUBJ-09	Pending Identification / Enrollment	79	1	2026-09-25 14:03:24	2026-09-25 14:15:57	Morning / Day	**0**
SUBJ-10	Pending Identification / Enrollment	75	2	2026-09-13 19:40:38	2026-09-21 19:37:46	Morning / Day	**0**

4. CRITICAL PLANT HAZARD ZONE COLLISION LEDGER

Plant Asset / Hazard Zone	Spatial Bounding Box [x1, y1, x2, y2]	Hazard Severity	Direct Contact	Proximity Visits	Operational Compliance Verdict
Two Multiport Backwash Valves	[755, 220, 985, 360]	**CRITICAL (Filtration)**	**0 Breaches**	0 Events	**100% UNTOUCHED / ZERO TAMPERING**
Ultimate Plant Power Switch	[998, 155, 1090, 265]	**SEVERE (Main Breaker)**	**0 Breaches**	0 Events	**100% UNTOUCHED / UNINTERRUPTED POWER**
RO Skid Controls & Walkway	[945, 345, 1210, 510]	**ELEVATED (Monitoring)**	Authorized Only	Routine Operations	**COMPLIANT OPERATIONAL ACCESS**

FORENSIC INTEGRITY CERTIFICATION:
- Master Surveillance Source: E:\RO WATER SEP (Mounted in strictly Read-Only custody :ro; zero drive modification).
- All 999 master 3K clips screened across 30 continuous calendar days (713.66 recorded hours).
- Persona classifications remain strictly objective (SUBJ-01 to SUBJ-XX) pending formal identity enrollment.
- Hash verification and pgvector 512-D embedding matrices synced with Supabase PostgreSQL.